

Netflix Recommendation System using Collaborative Filtering and Matrix Factorization (SVD)

1. Introduction

Recommendation systems have become an essential component of modern digital platforms such as Netflix, Amazon, YouTube, and Spotify. These systems analyze user preferences and historical interactions to suggest relevant content, thereby improving user engagement and satisfaction.

The objective of this project is to develop a movie recommendation system using Collaborative Filtering techniques and compare the performance of Item-Based Collaborative Filtering (Item-CF) with Singular Value Decomposition (SVD), a Matrix Factorization approach. The project also includes exploratory data analysis, model evaluation, and deployment through an interactive Streamlit dashboard.

2. Problem Statement

With the rapid growth of online content, users often face difficulty in discovering movies that match their interests. Traditional search methods are insufficient for providing personalized recommendations.

The goal of this project is to build a recommendation system capable of:

- Learning user preferences from historical ratings.
- Predicting unseen movie ratings.
- Recommending relevant movies to users.
- Comparing different recommendation algorithms.
- Providing an interactive dashboard for visualization and recommendation generation.

3. Objectives

The primary objectives of this project are:

1. Perform exploratory analysis of the Netflix dataset.
 2. Build an Item-Based Collaborative Filtering model.
 3. Build a Matrix Factorization model using SVD.
 4. Evaluate models using RMSE, MAE, and MAP@10.
 5. Compare recommendation quality across models.
 6. Develop an interactive dashboard using Streamlit.
 7. Generate personalized movie recommendations.
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4. Dataset Description

The project utilizes a subset of the Netflix Prize Dataset.

Dataset Statistics

- Total Movies: 225
- Total Users: 283,705
- Total Ratings: 1,000,000

Features Used

Feature Description

UserID Unique identifier for each user

MovieID Unique identifier for each movie

Rating User rating for a movie

Date Rating timestamp

Title Movie title

Feature Description

Year Release year

The dataset contains user-movie interactions which are used to learn viewing preferences and generate recommendations.

5. Data Preprocessing

Before model development, the dataset underwent several preprocessing steps:

Data Cleaning

- Removed duplicate records.
- Checked for missing values.
- Standardized column formats.

Data Transformation

- Merged movie metadata with rating information.
- Converted rating information into user-item interaction matrices.
- Prepared sparse matrices suitable for collaborative filtering.

Feature Engineering

- Generated user activity statistics.
- Computed movie popularity measures.
- Created user-movie interaction matrices.

These preprocessing steps improved data quality and model performance.

6. Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to understand rating behavior and dataset characteristics.

Rating Distribution

The majority of ratings fall between 3 and 5 stars, indicating generally positive user feedback.

User Activity Analysis

A small group of users contributes a large number of ratings, while most users rate relatively few movies.

Movie Popularity Analysis

Certain movies receive significantly higher numbers of ratings, indicating popularity imbalance.

Temporal Analysis

Movie rating activity increases over time, demonstrating growing platform engagement.

Interaction Matrix Visualization

A user-movie interaction heatmap was created to illustrate the sparsity characteristic commonly found in recommendation datasets.

7. Methodology

7.1 Item-Based Collaborative Filtering

Item-Based Collaborative Filtering recommends movies by identifying similarities between items.

Working Principle

1. Construct a user-item rating matrix.
2. Compute item-to-item similarity.
3. Identify movies similar to previously liked movies.
4. Generate recommendations based on similarity scores.

Advantages

- Simple and interpretable.
- Effective for dense datasets.

- Easy to implement.

Limitations

- Suffers from sparsity issues.
 - Limited scalability for large datasets.
 - Difficulty handling unseen interactions.
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7.2 Singular Value Decomposition (SVD)

SVD is a Matrix Factorization technique that decomposes the user-item interaction matrix into latent feature representations.

Working Principle

The user-item matrix is decomposed into:

$$R = U\Sigma V^T$$

Where:

- U = User latent factors
- Σ = Singular values
- V^T = Movie latent factors

The model predicts unknown ratings by learning hidden relationships between users and movies.

Advantages

- Handles sparse data effectively.
- Learns latent user preferences.
- Produces high-quality recommendations.
- Better scalability than neighborhood methods.

Limitations

- Computationally intensive.
- Requires model training.

- Less interpretable than similarity-based methods.

8. Evaluation Metrics

The recommendation models were evaluated using three performance metrics.

Root Mean Square Error (RMSE)

RMSE measures prediction accuracy by penalizing large prediction errors.

Lower RMSE indicates better performance.

Mean Absolute Error (MAE)

MAE calculates the average absolute difference between predicted and actual ratings.

Lower MAE indicates more accurate predictions.

Mean Average Precision at 10 (MAP@10)

MAP@10 evaluates ranking quality by measuring how effectively relevant movies appear among the top recommendations.

Higher MAP@10 indicates better recommendation quality.

9. Experimental Results

Model Performance Comparison

Metric	Item-Based CF	SVD
RMSE	1.1421	0.9924
MAE	0.8870	0.7775
MAP@10	0.9372	0.9512

Analysis

- SVD achieved the lowest RMSE.
- SVD achieved the lowest MAE.
- SVD achieved the highest MAP@10.
- SVD consistently outperformed Item-Based Collaborative Filtering across all evaluation metrics.

Therefore, SVD was selected as the best-performing recommendation model.

10. Dashboard Development

A Streamlit-based interactive dashboard was developed to visualize dataset insights and recommendation results.

Dashboard Modules

Overview

- Project summary
- Dataset statistics
- Project objectives

Movie Explorer

- Movie search and selection
- Movie metadata display

Analytics

- Rating distributions
- Popularity analysis
- User activity analysis
- Interaction matrix visualization

Recommendation Engine

- User ID input

- Top-10 personalized recommendations
- Recommendation generation using SVD

Model Results

- Performance comparison charts
- Evaluation metrics

About Project

- Technologies used
- Algorithms implemented
- Key findings

11. Technologies Used

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Scikit-Learn
- Surprise Library
- Streamlit

Deployment Tools

- Streamlit
 - Ngrok
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12. Challenges Faced

Several challenges were encountered during project development:

1. Handling sparse user-item matrices.
2. Managing large-scale rating data.
3. Selecting suitable evaluation metrics.
4. Training matrix factorization models efficiently.
5. Integrating recommendation results into an interactive dashboard.
6. Deploying the application for remote access using Ngrok.

These challenges were addressed through preprocessing, matrix factorization techniques, and dashboard optimization.

13. Key Findings

The major findings from the project are:

- SVD significantly outperformed Item-Based Collaborative Filtering.
 - Matrix Factorization effectively captures latent user preferences.
 - Recommendation quality improved substantially using SVD.
 - Interactive dashboards enhance usability and project presentation.
 - Collaborative filtering remains an effective approach for personalized recommendations.
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14. Future Work

Future enhancements may include:

- Hybrid recommendation systems.
- Content-based filtering integration.
- Genre-aware recommendations.
- Cold-start user handling.

- Deep Learning-based recommendation models.
 - FastAPI deployment.
 - Cloud deployment using AWS or Azure.
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15. Conclusion

This project successfully developed a Netflix Movie Recommendation System using Collaborative Filtering and Matrix Factorization techniques. Two recommendation approaches, Item-Based Collaborative Filtering and SVD, were implemented and evaluated using RMSE, MAE, and MAP@10 metrics.

Experimental results demonstrated that SVD consistently outperformed Item-Based Collaborative Filtering, achieving lower prediction errors and higher recommendation quality. The final recommendation engine was integrated into a Streamlit dashboard that enables users to explore movies, analyze dataset characteristics, compare model performance, and generate personalized recommendations.

The project demonstrates the effectiveness of recommendation systems in enhancing user experience and highlights the importance of Matrix Factorization techniques in modern recommender systems.

DEMO VIDEO-

<https://drive.google.com/drive/folders/18GiYFSL7UHJbxaaGxKrUNCIHhD54UUgX?usp=sharing>